

THE BINARY NUMBER SYSTEM

CSE 1110: Introduction to Computer Systems

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WHY BINARY?

Major computer components (CPU, RAM etc.) are made of transistors

A transistor is an electronic switch

- Has two states – on (1) and off (0)

This is why a computer understands the language of 1's and 0's only

- Binary number system

Every instruction of a computer is **encoded** in binary

Every data is also **encoded** in binary

Computer stores and manipulates these data in binary format

BINARY NUMBER SYSTEM

Two digits to represent every number – 0 (zero) and 1 (one)

- Known as bits

8 bits together – a byte

A number of bytes together – a word

- Usually 4 or 8 bytes, depending on the processor, computer architecture etc.

More units – kilobytes, megabytes, gigabytes, terabytes, ...

0	0
1	1
2	10
3	11
4	100
5	101
6	110
7	111
8	1000
9	1001
10	1010
11	1011
12	1100
13	1101
14	1110
15	1111
16	10000

COMPUTER STORES DATA IN ITS BINARY FORM

Let's see how a number is stored in its binary format

PART 1: CONVERT A DECIMAL INTEGER TO BINARY

1. Divide the number by 2
2. Store the quotient and remainder
3. Let the quotient be the new number
4. Repeat steps 1 to 3 until the number is 0
5. Write down the remainders from last to first. That will be the binary format

EXAMPLE: CONVERT $(154)_{10}$ TO BINARY

	<u>Quotient</u>	<u>Remainder</u>
2	154	
2	77	0
2	38	1
2	19	0
2	9	1
2	4	1
2	2	0
2	1	0
	0	1

$(154)_{10} = (10011010)_2$

PART 2: CONVERT A PROPER FRACTION TO BINARY

1. Multiply the number by 2
2. Separate the integer and the fraction parts
3. Let the fraction part be the new number
4. Repeat steps 1 to 3 until the number is 0
5. Write down the integer parts from first to last after the binary point. This will be the binary representation

EXAMPLE: CONVERT $(0.625)_{10}$ TO BINARY

<u>Integer</u>	<u>Fraction</u>
-	.625 x2
1	.250 x2
0	.500 x2
1	.000



$$(0.625)_{10} = (0.101)_2$$

HOW TO CONVERT IMPROPER FRACTIONS?

To convert $(154.625)_{10}$ to binary –

- Divide the number into integer and fraction part, i.e. $(154)_{10}$ and $(0.625)_{10}$
- Convert them individually to binary
- Join them up

$$(154.625)_{10} = (10011010.101)_2$$

REVERSE ACTION: BINARY TO DECIMAL

Multiply each bit with the place value of its place value, and sum the results

Place values –

2^6	2^5	2^4	2^3	2^2	2^1	2^0	.	2^{-1}	2^{-2}	2^{-3}	2^{-4}
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REVERSE ACTION: BINARY TO DECIMAL

$$(10011010.101)_2$$

$$= 1 \times 2^7 + 0 \times 2^6 + 0 \times 2^5 + 1 \times 2^4 + 1 \times 2^3 + 0 \times 2^2 + 1 \times 2^1 + 0 \times 2^0 + 1 \times 2^{-1} + 0 \times 2^{-2} + 1 \times 2^{-3}$$

$$= 128 + 0 + 0 + 16 + 8 + 0 + 2 + 0 + 0.5 + 0 + 0.125$$

$$= (154.625)_{10}$$

COMPUTER MANIPULATES DATA IN ITS BINARY FORM

Let's see a basic manipulation example

ADDING TWO BINARY NUMBERS

$$0101 + 0111 = 1100$$

$$0101 + 1011 = 10000$$

$$0111 + 0011 = 1010$$

How is it done?

- Watch carefully ...

ADDING TWO BINARY NUMBERS

1101011
+100110
=====
10010001

1 1 1 1 1
1 1 0 1 0 1 1
+ 1 0 0 1 1 0

1 0 0 1 0 0 0 1

IMPLEMENTING THE MANIPULATIONS IN CIRCUITS: BOOLEAN OPERATIONS

Every manipulation is a combination of three basic operations

- Called the Boolean operations, by the name of George Bool

Three Boolean operations

- Logical addition (OR)
- Logical multiplication (AND)
- Inversion (NOT)

These operations are implemented using logic gates

- Gates take one or more bits as inputs, and give one bit an output, according to the operations